

View results

Respondent

31

Anonymous

52:28

Time to complete

A. General information about the respondent

1. * Country

Slovenia



2. * Name of organization(s) or entity(s) responsible for the preparation of the report

Ministry of Higher Education, Science and Innovation, UNESCO Office

3. * Website of organization(s) or entity(s) responsible for the preparation of the report

<https://www.gov.si/en/state-authorities/ministries/ministry-of-higher-education-science-and-innovation/>

4. * Email address of organization(s) or entity(s) responsible for the preparation of the report

5. Phone number of organization(s) or entity(s) responsible for the preparation of the report

6. Please describe the role/mandate of your organization

The Ministry of Higher Education, Science and Innovation focuses on the quality development of higher education, the transfer of knowledge into practice for the development of society as a whole, and the creation of a stimulating environment for research and innovation

7. * Full name of officially designated contact person(s) who completed this survey

Peter Sterle

8. * Position of officially designated contact person(s) who completed this survey

Secretary

9. * Email address of officially designated contact person(s) who completed this survey

10. Full Name(s) of designated official(s) certifying the report

Ivan Skubic

11. Brief description of the consultation process established for the preparation of the report

The Ministry for Higher Education, Science and Innovation consulted the Slovenian Open Science Community, which formed a working group that contributed responses to the questions below. The members of the group were (in alphabetical order): Maja Bogataj Jančič (IPI), Marko Drobnjak (Arnes), Mojca Kotar (UL), Dunja Legat (UKM), Aleš Pogačnik (ZRC SAZU), Miro Pušnik (CTK), Tea Romih (CTK), Ana Slavec (UP) and Ivan Skubic (MVZI).

12. * Name of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

Slovenian Open Science Community

13. * Website of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey
(Please use commas to separate multiple links, as: link A, link B)

<https://odprtaznanost.si/>

14. * Email address of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

15. Sector of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey



Public



Private



Civil Society



Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING

ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

Part 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,

ASSOCIATED BENEFITS AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN
SCIENCE

16. 1.1 Has the 2021 UNESCO Recommendation on Open Science been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?



Yes



No

17. 1.1 cont. **If yes**, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

The recommendation has been shared with the Ministry of Higher Education, Science and Innovation, which has promoted it among different stakeholders. The Recommendation was promoted together with EU strategic documents and recommendations in the area of open science, which are compatible with the 2021 UNESCO Recommendation. In 2020, the Ministry of Higher Education, Science and Innovation set up a national Working group, consisting of all the relevant stakeholders, whose task was to prepare a National research and innovation strategy involving open science as well as an Action Plan for Open Science. The role of these documents was to enable implementation of international open science policies (UNESCO, EU/ERA) into the national system.

18. 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

☐ Yes

☒ No

19. 1.3 Have there been awareness raising activities on open science, including all its key elements^[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (*ref.: (i) 16.f, 16.g, 16.h*)

[i] Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

☒ Yes

☐ No

20. 1.3 cont. **If yes**, please provide details on the activities organized or foreseen and links as relevant.

Slovenia has been actively organising and planning various initiatives to raise awareness about open science, encompassing its key elements, associated benefits and challenges. In 2022, 2023 and 2024, the Academic and Research Network of Slovenia (Arnes), the Slovenian Open Science Community (SSOZ), and the Ministry of Higher Education, Science and Innovation have jointly organised Open Science Days and EOSC National Tripartite Event, and plan to organise this event in 2025 as well. From the systemic point of view the awareness rising is also embedded into the national project SPOZNAJ (2023-2026), financed by the EU/RRF and the national budget. Additionally, several promotion and awareness-raising activities are organised within the Slovenian Citizen Science Network.

21. 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[i], in publicly funded research? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.

Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

☒ Yes

☐ No

22. 1.4 cont. If **yes**, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

Slovenia has undertaken several actions to incorporate the values and principles of open science into publicly funded research. In December 2021, it enacted the Scientific Research and Innovation Activities Act (ZZrID), which mandates open access to research results funded by public resources and emphasizes the involvement of the public in scientific research and the evaluation of research activities in line with open science principles. In March 2022, it adopted the Resolution on the Slovenian Scientific Research and Innovation Strategy 2030 (ReZrIS30), which identifies open science as a central theme, aiming to improve the quality of research. In May 2023, it adopted the Decree on the Implementation of Scientific and Research Work and the Action Plan for Open Science, which outline specific activities, indicators, responsible entities, timelines, and necessary resources to effectively implement open science principles.

Part 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

23. 2.1 Does your country have a national policy[i], strategy or plan of action on science, technology and innovation (STI)?

[i] National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

☒ Yes

☐ No

24. 2.1 cont. **If yes**

- please share a machine-readable PDF file of the policy using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

In the comment box, please indicate the question number for this attachment / document.

After uploading, please return to this page to continue the survey.

National policy documents that involve important Open science provisions are Slovenian Act on Scientific Research and Innovation Activities (ZZrID) <https://view.officeapps.live.com/op/view.aspx?src=https%3A%2F%2Fwww.gov.si%2Fassets%2Fministrstva%2FMVZI%2FZnanost%2FNacionalne-strategije-in-dokumenti%2FScientific-Research-and-Innovation-Activities-Act.doc> (adopted by National Assembly on 18 November 2021, in force on 1 January 2022, last amended on 18 November 2024, covering - in particular Articles 40, 41 and 42 - open access to research results, evaluation of the scientific research work to promote open science and to use responsible metrics, involvement of interested public including citizen science into the research process); Resolution on the Slovenian Scientific Research and Innovation Strategy 2030 (ReZrIS30) <https://www.gov.si/assets/ministrstva/MIZS/Dokumenti/ZNANOST/Nacionalni-dokumenti/Resolution-on-the-Slovenian-Scientific-Research-and-Innovation-Strategy-2030> (adopted by National Assembly on 23. March 2022. Measure 6.2. is covering management and financing of national open science ecosystem including the open science infrastructure, introduction of modern approaches to the evaluation of scientific research (DORA, CoARA,...), FAIR research results, establishment of National Open Science Community, promotion and development of Citizen Science and public involvement in scientific research, promotion and development of national open access scientific publishing); Action Plan for Open Science <https://www.gov.si/assets/ministrstva/MVZI/Znanost/Nacionalne-strategije-in-dokumenti/2023-Action-Plan-Open-Science-Slovenia.pdf> (adopted by the Government of the Republic of Slovenia, on 31. May 2023, covering several actions directly related to the implementation of the provisions of ReZrIS30; Decree on the implementation of scientific research work in accordance with the principles of open science <https://www.gov.si/assets/ministrstva/MVZI/Znanost/Obzorje-Evropa/Novice/2023-Open-Science/Decree-on-the-implementation-of-scientific-research-work-open-science.pdf> (adopted by the Government of the Republic of Slovenia on 25. May 2023, detailing the open science provisions of ZZrID Act)

25. 2.1 cont. **If yes:**

- does it include any of the following open science elements:

- ☒ open access to scientific knowledge
- ☒ open science infrastructure
- ☒ open engagement of societal actors
- ☒ open dialogue with other knowledge systems

26. 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[i] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

- ☐ Yes
- ☒ Partly
- ☐ No

27. 2.2 cont. **If yes or partly**, please share a machine-readable PDF file(s) of the policy(ies) using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

Recommendation (16a) is introduced through many of the activities of the Action Plan for Open Science <https://www.gov.si/assets/ministrstva/MVZI/Znanost/Nacionalne-strategije-in-dokumenti/2023-Action-Plan-Open-Science-Slovenia.pdf> and the activities of Slovenian Open Science Society <https://odprtaznanost.si>. Recommendation (16b) is introduced through the governmental decree on open science <https://www.gov.si/assets/ministrstva/MVZI/Znanost/Obzorje-Evropa/Novice/2023-Open-Science/Decree-on-the-implementation-of-scientific-research-work-open-science.pdf>, in particular through Article 6 and Article 7. Recommendation (16c) is implemented through the Resolution on the Slovenian Scientific Research and Innovation Strategy 2030 <https://www.gov.si/assets/ministrstva/MIZS/Dokumenti/ZNANOST/Nacionalni-dokumenti/Resolution-on-the-Slovenian-Scientific-Research-and-Innovation-Strategy-2030>, in particular through Measure 6.2.6. and the related Action plan for Open science and the Actions under Measure 6.2.6. that are related to the promotion and development of national scientific publishing that will operate according to the principles of open science. The governmental Decree on Open Science is addressing the bibliodiversity in Article 1, paragraph (2). Transitional provisions (Arts. 14 and 15) are related to bibliodiversity as well. Recommendation (16d) is not specifically addressed. Recommendation (16h) is enabled and supported through (1): the National open science community (Slovenian Open Science Community) which is the result of Measure 6.2.4 of the Resolution on the Slovenian Scientific Research and Innovation Strategy 2030 and related activities of the Action plan for Open Science under M.6.2.4. and through (2): the promotion and development of citizen science and public involvement in scientific research which is the result of Measure 6.2.5 of Resolution on the Slovenian Scientific Research and Innovation Strategy 2030 and related activities of the Action plan for Open Science under M.6.2.5.

28. 2.3 In your country, are there specific policy instruments[i] that aim to promote open science?

[i] Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).



Yes



Under development



No

29. 2.3 cont. **If yes or under development**, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

The main policy instrument is the Action plan for Open Science. The Slovenian Research and Innovation Agency, the main funder of Slovenian science, includes open science requirements in its calls for proposals in accordance with the 2021 Act. Besides there are the regular annual calls for proposals for scientific journals and monographs where the journal or book need to be in immediate open access. Additionally, the SPOZNAJ (2023-2026) introduces open science principles to Slovenian public research organisations.

30. 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)



Yes



Under development



No

31. 2.4 cont. **If yes or under development**, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

Since all EU/ERA recommendations and policies in the field of open science were developed in alignment with the UNESCO recommendations and are fully consistent with them, all Slovenian national documents that were prepared by the Ministry for Higher Education, Science and Innovation are also in line with the listed UNESCO recommendations (details are under item 2.1 of this document).

32. 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (*ref.: (ii) 17.d, 17.e, 17.g*)



Yes



Under development



No

33. 2.5 cont. **If yes**, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors). To view the elements of open science,

see: <https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/page=9>

First, the institutional implementation of the principles of open science in the activities of public research organisations takes place through the Open Science Action Plan (<https://www.gov.si/assets/ministrstva/MVZI/Znanost/Nacionalne-strategije-in-dokumenti/2023-Action-Plan-Open-Science-Slovenia.pdf>), which introduces a set of integrated infrastructures and activities that support the adaptation of research organisations to operate according to open science principles, ranging from the provision of open science infrastructure and open science monitoring to the provision of skills and competences. Second, the project SPOZNAJ (2023-2026) (<https://projekt-spoznaj.si>), which is funded by the Ministry of Higher Education, Science and Innovation and the European Union – NextGenerationEU, aims to integrate the principles of open science into scientific research practices and to align the operations of twenty public research organisations and the Central Technical Library at the University of Ljubljana with these principles. In particular, Work Package 6 of the SPOZNAJ project is dedicated to the comprehensive adjustment of consortium partners' operations in accordance with the open science principles and evaluation of the results. It is divided into the following subsets: Subset 6.1 Analysis of business/organizational models of support structures to facilitate operations in accordance with open science principles, tailored to the specific needs of individual consortium partners and providing recommendations based on the findings. The project partners will analyze their organization's situation regarding research data management and legal, ethical, and other aspects related to open science. Subset 6.2 Plan for operational adjustments in accordance with open science principles and piloting the implementation of the most significant adjustments. In the analysis, the project partners will identify necessary changes to internal legal acts and organizational structures required to enforce open science principles. They will also assess the presence of suitable infrastructures and their integration. Subset 6.3 Preparation of internal organizational regulations, documents, and policies for the final recipients. The project partners will incorporate provisions on open science into their internal regulations, which govern the implementation of stable funding for research activities. Subset 6.4 Making adjustments. Simultaneously with the preparation of legal acts outlined in subset 6.4, the project partners will undertake organizational, personnel, and infrastructural adaptations of work processes, technologies, digital repositories, and libraries. Subset 6.5: Evaluation of project results. The project partners will verify the achievement of the planned project goals through an evaluation process that will consider the requirements of the public tender and the established objectives. Additionally, at the University of Ljubljana open science is supported by the Statutes of UL (<http://pisrs.si/Pis.web/pregledPredpisa?id=STAT233>) Art. 6 and 133, Rules on stable funding of UL scientific research activities (<https://www.uni-lj.si/assets/Pravni-akti/Raziskovalno-razvojno-delo-in-intelektualna-lastnina/Trenutno-veljavni-akti-UL/Raziskovalno-in-razvojno-delo/Pravilnik-o-stabilnem-financiranju-znanstvenoraziskovalne-dejavnosti-UL-neuradno-precisceno-besedilo-velja-od-6.4.2024.pdf>), Art. 25 and 26, Rules on doctoral studies at UL (<https://www.uni-lj.si/assets/Sluzba-za-doktorski-studij/Pravila-in-postopki/Pravilnik-o-doktorskem-studiju-neuradno-precisceno-besedilo-z-dne-27.-4.-2024.pdf>), mostly Art. 50, Strategy of UL 2022-2027 (<https://www.uni-lj.si/assets/Sluzba-za-spremljanje-kakovosti-analize-in-porocanje/Strategija-UL->

2022-2027.pdf), Item 7.2, performance indicator RP2-K6m, and Digital strategy of UL 2024-2027 (https://www.uni-lj.si/assets/Kabinet/1_Strategije-UL/08_Digitalna_strategija_UL_2024-2027.pdf), Goal Research R.2 and activity 2. At

34. 2.6 In your country, are there specific funding mechanisms[i] for open science at the national level? (*ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c*)

[i] A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.



Yes



Under development



No

35. 2.6 cont. **If yes or under development**, please provide details and links as relevant.

The main funders of scientific research in Slovenia are the Ministry for Education, Science and Innovation and the Slovenian Research and Innovation Agency. The first provides resources for open science through funding activities within the Action Plan for Open Science and co-funding project SPOZNAJ (2023-2026), while the second includes a Sector of Open Science, which covers the following public tenders: 4.1 Public tender for (co)financing of research equipment purchases <https://www.aris-rs.si/sl/infra/oprema/razpisi/23/razp-oprema-23.asp> 4.2 Public tender for (co)financing of scientific monographs <https://www.aris-rs.si/sl/razpisi/24/razp-monografije-24.asp> 4.3 Public call for (co)financing of the publication of domestic periodical scientific publications <https://www.aris-rs.si/sl/infra/tisk/razpisi/24/razp-zpp-25-26.asp> 4.4 Public Call for Co-financing of the Publication of Domestic Periodical Popular Science Publications <https://www.aris-rs.si/sl/infra/tisk/razpisi/24/razp-ppp-25.asp> 4.5 Public Call for Co-financing of the Purchase of International Scientific Literature (MZL) <https://www.aris-rs.si/sl/infra/tujlit/razpisi/24/razp-tujlit-24.asp> 4.6 Public Call for (co)financing of science popularization activities <https://www.aris-rs.si/sl/razpisi/24/razp-popular-znanosti-24.asp> 4.7 Public call for co-financing of activity programs of central specialized information centers for research activities <https://www.aris-rs.si/sl/infra/osic/razpisi/22/razp-OSIC-23-25.asp> 4.8 Public call for reimbursement of costs of scientific publications in open access <https://www.aris-rs.si/sl/razpisi/24/razp-odprti-dostop-24.asp> 4.9 Call for collection of data on the involvement of members of program groups in projects outside the ARRS funding framework – A3 <https://www.aris-rs.si/sl/obvestila/24/poziv-a3-2024.asp> Additionally, in 2024 the agency introduced a Public call for (co)financing of citizen science activities. <https://www.aris-rs.si/sl/dostop/razpisi/24/razpis-obcanska-znanost-24.asp>

36. 2.7 1.1 In your country, have open science practices[i] been integrated in existing research funding mechanisms?

[i] Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

☒ Yes

☐ No

37. 2.7 cont. **If yes**, please provide details and links as relevant.

In accordance with Article 41 of the Scientific Research and Innovation Act and Articles 2 and 3 of the Regulation on the Implementation of Scientific Research Work in Accordance with the Principles of Open Science, all peer-reviewed scientific publications resulting from the implementation of research projects funded by ARIS, together with the associated research data and other research results, must be published in open access. From 2025, the agency requires project leaders to submit a Data Management Plan and Communication Plan up to six months after the start of the project.

38. 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

☐ Yes

☒ No

39. 2.8 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

/

40. 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.j, 23.c)

☐ Yes

☒ Under development

☐ No

41. 2.9 cont. **If yes or under development**, please provide details and links as relevant.

We are performing the monitoring for open science by two mechanisms. The first mechanism are annual work plans and annual reports of publicly funded research organisations (we have 22 publicly funded research organisations in Slovenia). In these documents they are, besides other things, planning and reporting (mostly in a narrative way) their progress in aligning operations towards open science principles as stated in Resolution on the Slovenian Scientific Research and Innovation Strategy 2030, Decree on the implementation of scientific research work in accordance with the principles of open science (<https://www.gov.si/assets/ministrstva/MIZS/Dokumenti/ZNANOST/Nacionalni-dokumenti/Resolution-on-the-Slovenian-Scientific-Research-and-Innovation-Strategy-2030>) and pertaining Action plan for Open Science. In doing so they are supported by a SPOZNAJ (2023-2026) (<https://projekt-spoznaj.si>) project. The second mechanism is a national open science monitor which displays numerical data of the level of openness of different research results in digital form. It is currently being piloted (<https://monitor.openscience.si/>). The data for the national monitor is mainly based on data from the national research information system SICRIS, the national bibliographic system COBISS, the Slovenian Open Science Infrastructure and the national Open Data Portal. In the future, we intend to improve the quality of the data by adding data from some international data sources such as Crossref, Open Alex, Datacite, Directory of Open Access Journals, etc. We expect the Slovenian Open Science Monitor to be fully operational in the second half of 2025.

Part 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

42. 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

According to the Statistical Office of the Republic of Slovenia, the latest official data indicate that in 2022, Slovenia's gross domestic expenditure on research and development (R&D) was 2.10% of its GDP (SORS, March 2024) (<https://www.stat.si/StatWeb/en/News/Index/12716>).

43. 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

According to the Statistical Office of the Republic of Slovenia, in 2023, 94% of households with at least one resident aged 16–74 had internet access at home (SORS 2024 <https://www.stat.si/StatWeb/en/News/Index/13153>). At the end of the 1st quarter of 2024, 683.873 optical internet connections were used in Slovenia (SORS 2024 <https://www.stat.si/StatWeb/en/news/Index/12929>).

44. 3.3 Are there national research and education networks (NRENs)[i] active in the country? (ref.: (iii) 18.c)

[i] A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

- ☒ Yes
- ☐ No

45. 3.3 cont. **If yes**, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

The Academic and Research Network of Slovenia (ARNES <https://www.arnes.si/en/about-arnes>), established in 1992, plays a crucial role in advancing research, education, and culture by providing institutions and their users with cutting-edge technological opportunities. Its rapid expansion within universities, libraries, and schools significantly contributed to the widespread adoption of the internet in Slovenia, connecting students, educators, and researchers. Today, ARNES manages the country's scientific research infrastructure, encompassing backbone network infrastructure, data centers, supercomputing clusters, and network services.

46. 3.4 Are there national or institutional open science infrastructures[i] in your country? (ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)

[i] Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.



Yes



Under development



No

47. 3.4 cont. **If yes or under development**, please indicate which type(s) of infrastructure:



federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage



community managed infrastructures, protocols and standards, for example those that support biodiversity and engagement with society



platforms for exchanges and co-creation of knowledge between scientists and society



community-based monitoring and information systems to complement national, regional and global data and information systems

48. 3.4 cont. **If yes or under development**, for each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

Slovenia has two public infrastructural institutions responsible for the development and management of key research infrastructure. ARNES (Academic and Research Network of Slovenia <https://www.arnes.si/en/about-arnes/>) provides research, educational, and cultural institutions with access to advanced technologies and coordinates the Slovenian National Supercomputing Network (SLING). In addition to offering network services, it manages the backbone network, data centers to support open science, and a supercomputing cluster. IZUM (Institute of Information Science <https://izum.si/en/aboutus/>) supports science, education, and culture by providing essential information infrastructure. Its key responsibilities include managing Vega, one of the most powerful supercomputers in the region, as well as developing and maintaining the COBISS.SI library information system. Both institutions play a crucial role in advancing digital infrastructure and supporting scientific research in Slovenia. community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society In Slovenia, various research infrastructures, protocols, and standards support open science, bibliodiversity, and societal engagement in different ways. While the exact definition of community-managed infrastructure is not always clear, several organizations actively involve researchers and the broader community in the development and use of research resources. Key research infrastructures in this context include CLARIN.SI, the Social Science Data Archive (ADP), ELIXIR Slovenia, and DARIAH-SI. CLARIN.SI (the Slovenian Research Infrastructure for Language Resources and Technologies <http://CLARIN.SI>) provides long-term archiving of language resources and ensures open and free access to these resources, tools, and services for researchers. While it is a formally managed infrastructure, it promotes linguistic diversity and engages the broader research community in linguistic studies. Social Science Data Archives (ADP <https://www.adp.fdv.uni-lj.si/eng/>) serves as the national data center for research data in the social sciences and humanities. It provides researchers with access to high-quality data resources and supports open science and data sharing, contributing to greater bibliodiversity and societal engagement. ELIXIR Slovenia <https://elixir-slovenia.org/> is the national node of the European research infrastructure for life sciences. It coordinates, connects, and maintains bioinformatics resources while providing access to key data, tools, standards, and training programs in bioinformatics. This fosters research and enables broader community participation in scientific activities. DARIAH-SI (Digital Research Infrastructure for the Arts and Humanities – Slovenia <http://www.dariah.si/en/>) is part of the European DARIAH network, supporting digital research in the arts and humanities. It facilitates access to digital tools, data, and knowledge, while also fostering collaboration among researchers, cultural institutions, and the wider public. By promoting open and interoperable standards, DARIAH-SI strengthens the digital humanities community and contributes to bibliodiversity. platforms for exchanges and co-creation of knowledge between scientists and society community-based monitoring and information systems to complement national, regional and global data and information systems. For each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

49. 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)



Yes



No

50. 3.5. cont. **If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open sciencei, stage of research, accessibility to all the relevant users in the region or internationally).

Slovenia actively participates in and supports several federated regional and international information technology infrastructures that advance open science. These infrastructures provide network connectivity, data-sharing frameworks, computational resources, and services that facilitate research collaboration at both national and international levels. All infrastructures operate in both Slovenian and English language. Some key initiatives include: The Academic and Research Network of Slovenia (ARNES) was established in 1992 and provides high-speed network infrastructure and digital services for the research and education community in Slovenia. As part of GÉANT, the pan-European research and education network, it ensures connectivity for Slovenian researchers and institutions, facilitating cross-border scientific collaboration. Target groups and beneficiaries: Researchers, educators, students, cultural institutions, public administration Governance/community agreements: Public institution governed by the Ministry of Higher Education, Science, and Innovation, member of GÉANT Scope: Network infrastructure, digital identity services, cloud services, e-learning, and security Accessibility: Available to research and educational institutions in Slovenia, with international collaboration through GÉANT Vega Supercomputer (EuroHPC) was established in 2021. Slovenia is a member of the EuroHPC Joint Undertaking, a European initiative for high-performance computing. The country hosts the Vega supercomputer, operated by IZUM, which is part of the EuroHPC network and provides computational power to researchers across Europe. Target groups and beneficiaries: Researchers in computational sciences, engineering, artificial intelligence, climate modelling, etc. Governance/community agreements: Managed by IZUM, part of the EuroHPC Joint Undertaking, co-funded by the European Commission Scope: High-performance computing (HPC), AI, data analytics Accessibility: Open to researchers in Slovenia and Europe through EuroHPC Access Calls European Open Science Cloud (EOSC) to which Slovenia contributes through its mandated organization, ARNES, which is set to become the national EOSC EU Node in 2026 or 2026. This initiative supports the integration and accessibility of open research data and services. Target groups and beneficiaries: Researchers, institutions, policymakers, industry Governance/community agreements: Coordinated by ARNES and the Slovenian Community of Open Science, aligned with the EOSC governance structure Scope: Open science data sharing, FAIR principles, research data management Accessibility: Open to researchers and institutions in Slovenia and integrated into the broader European EOSC framework Slovenia actively participates in European data-sharing infrastructures. The Social Science Data Archive (ADP) was established in 1997 as the national representative in CESSDA ERIC, ensuring long-term data preservation and access in the social sciences. Additionally, CLARIN.SI (for linguistic resources) and DARIAH-SI (for digital humanities) contribute to European federations supporting open data and research collaboration. Target groups and beneficiaries: Social science researchers, policymakers, students Governance/community agreements: Member of CESSDA ERIC, governed by the Faculty of Social Sciences, University of Ljubljana Scope: Research data management, social science data preservation, FAIR data Accessibility: Open access and secure data-sharing policies for Slovenian and European researchers CLARIN.SI (Linguistic Resources) was established in 2015/2016. Target groups and beneficiaries: Linguists, digital humanities researchers, AI and NLP developers Governance/community agreements: Member of CLARIN ERIC, hosted at Jožef Stefan Institute (IJS) Scope: Language data repositories, computational linguistics, FAIR data principles Accessibility: Open to researchers and institutions in Slovenia and globally DARIAH-SI (Digital Humanities) was es

51. 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

☐ Yes

☒ No

Part 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY

AND CAPACITY BUILDING FOR OPEN SCIENCE

52. 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

☒ Yes

☐ No

53. 4.1 cont. **If yes**, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

Capacity building takes place through various channels, including universities, faculties and research-performing organizations. In addition, projects that focus on open science, such as SPOZNAJ <http://projekt-spoznaj.si/en/> and the National Action Plan for Open Science, offer various courses that deal with basic aspects of open science, open publishing, research data management, data and software publishing, authors' rights and more. Within this project, the first handbook on open science in the Slovenian language was published in 2024 (Spoznaj FAIR <https://www.cessda.eu/News/CESSDA-Newsitem-nid3939>). Moreover, at the University of Ljubljana, the Faculty of Social Sciences and some other faculties integrated open science topics into compulsory general lectures designed to prepare students for their doctoral theses. Additionally, the University of Maribor organized two international Open Science Summer Schools <https://osss.splet.arnes.si/2022-summer-school/> for doctoral and postdoctoral researchers, and programmes for university staff called IzpopolniUM <https://www.um.si/kakovost/izobrazevanja/>. In 2024, the University of Primorska started monthly open science seminars called OZUP <https://www.upr.si/si/o-univerzi/univerzitetna-knjiznica/novosti/>. Within the national action plan for implementing open science, part Skills and Competencies for Open Science, into the Slovenian research landscape, important tasks for three university libraries are foreseen. These are: organization of a network of information support for open science to provide coordinated training of researchers in support of FAIR and open science; establishment of professional support for addressing copyright and ethical issues in research work within the context of open science, and promotion of Slovenian science in the context of open science principles.

54. 4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

☐ Yes

☒ No

55. 4.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

There were no capacity-building programmes specifically organised for policymakers but they are regularly invited to open science events organised by different organisations and participate in working groups. They are well acquainted with open science policies by participating in the European research area open science activities. As a result they have integrated in a collaborative way core values and principles in the Slovenian legal background for research, i.e., in the Resolution on the Slovenian Scientific Research and Innovation Strategy 2030, Action Plan for Open Science, Research Infrastructure Roadmap 2030, Scientific Research and Innovation Activities Act, and Decree on the Implementation of Scientific Research Work in Accordance with the Principles of Open Science.

56. 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

☐

Yes

☒

No

57. 4.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

A framework of open science competencies has not yet been incorporated into higher education research skills curricula in Slovenia, and it is not foreseen by the end of 2025. Some capacity building is taking place for specific groups such as doctoral students on topics such as the preparation of data management plans and managing data in line with FAIR principles and "as open as possible" principles, and open publishing.

58. 4.4 Are open educational resources[i] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

[i] Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

- ☐ Yes
- ☒ Partly
- ☐ No

59. 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

- ☒ Yes
- ☐ No

60. 4.5 cont. **If yes**, please provide more information and links as relevant.

The central hub for information on open science events is the Slovenian Open Science Community <https://odprtaznanost.si/>, which brings together main Slovenian stakeholders as well as other (international) initiatives and institutions included in the Slovenian Action Plan for Open Science <https://www.gov.si/assets/ministrstva/MVZI/Znanost/Nacionalne-strategije-in-dokumenti/2023-Action-Plan-Open-Science-Slovenia.pdf>. Additionally, the portal <https://odprteobjave.si/> of the Central Technical Library covers international scientific publishing, the portal <https://odprtozaloznistvo.si/> of the University of Primorska covers domestic scientific publishing, and the ZRC SAZU manages and coordinates the activities within the national node OPERAS.SI <https://zalozba.zrc-sazu.si/sl/strani/operassi> of the European association OPERAS. The monthly ADP newsletter <https://www.adp.fdv.uni-lj.si/enovice/> is dedicated to events and education related to (social science) research data.

Part 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

61. 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: (v) 20.b, 20c, 20i)



Yes



No

62. 5.1 cont. **If yes**, please provide more information and links as relevant.

Yes, nine Slovenian organisations are members of the Coalition for Advancing Research Assessment (CoARA) <https://coara.eu/agreement/signatories/?category%5B0%5D=slovenia#signatories>, thus committing to reform the methods and processes by which research, researchers, and research organisations are evaluated. Moreover, the Slovenian Quality Assurance Agency for Higher Education (NAKVIS <https://nakvis.si/>) and the Slovenian Research and Innovation Agency (ARIS <https://www.aris-rs.si/sl/agencija/vizitka.asp>) have aligned minimal standards for election to the titles of higher education teachers and researchers with principles of open science (NAKVIS: Minimalni standardi za izvolitev v nazive visokošolskih učiteljev, znanstvenih delavcev in visokošolskih sodelavcev na visokošolskih zavodih, ARIS: Splošni akt o raziskovalnih nazivih <https://pisrs.si/pregledPredpisa?id=DRUG3538>). Public universities and research institutes are obliged by the Decree on the implementation of scientific research work in accordance with the principles of open science <https://www.gov.si/assets/ministrstva/MVZI/Znanost/Obzorje-Evropa/Novice/2023-Open-Science/Decree-on-the-implementation-of-scientific-research-work-open-science.pdf> (Arts. 10, 11, 17) to align assessment with open science principles. By doing so, they are supported by the SPOZNAJ project <http://projekt-spoznaj.si/en/>.

63. 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

☒ Yes

☐ No

64. 5.2 cont. **If yes**, please provide details and links as relevant.

One of the Slovenian signatories to CoARA (as well as DORA) is the Slovenian Research and Innovation Agency, the main funder of Slovenian science, which has announced that it will begin a complete overhaul of its evaluation system <https://www.aris-rs.si/sl/akti/24/prav-razisk-nazivi-dec24.asp> in May 2026. However, some institutions, such as the two largest Slovenian universities, the University of Ljubljana and the University of Maribor, have already started to change their assessment procedures in 2024 and have also organised several consultations on changes to the evaluation system.

65. 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)

☐ Yes

☒ No

66. 5.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Open science has only recently been included in the legal regulations in Slovenia. Within Measure 6.2.2/4 of the Action Plan for Open Science <https://www.gov.si/assets/ministrstva/MVZI/Znanost/Nacionalne-strategije-in-dokumenti/2023-Action-Plan-Open-Science-Slovenia.pdf> there are plans to introduce Incentives for Engagement in Open Science in the form of (1) amending internal regulations of JRO regarding the evaluation and assessment of researchers and (2) support for early-career researchers to encourage engagement in open science, including establishing a scheme to identify and recognize early-career researchers who have achieved notable results in the field of Open Science.

67. 5.4 Have there been any unintended negative consequences[i] of open science practices reported in your country? (ref.: (v) 20)

[i] For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.

☒ Yes

☐ No

68. 5.4 cont. **If yes**, please provide more information and specify if there is a strategy or plan to preventing and mitigating such consequences.

The negative effects can be reduced to a common denominator: The transition to open science is a technically and financially demanding endeavour that requires significant additional resources and efforts including cultural change. These factors are slowing down the changes.

Part 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

69. 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: (vi) 21)

- ☐ YES, at the national level
- ☐ YES, at the institutional level
- ☒ YES, at both the national and institutional levels
- ☐ No

70. 6.1 cont. **If yes**, please indicate which of the recommended actions they address:

- ☒ promotion of pre-prints
- ☒ exploration of open peer-review practices
- ☒ valuing and sharing of negative scientific results and associated data
- ☒ participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- ☒ development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas
- ☐ promoting open innovation practices
- ☐ Other

71. 6.1. cont. **If yes**, please provide more information and links as relevant

The Citizen Science Network <https://citizenscience.si/> is an active network in Slovenia that supports citizen science research and contributes to raising scientific literacy. The network currently has 55 members, ranging from universities to research institutions, NGOs and individuals. The network's catalogue currently includes 72 citizen science projects. In Slovenia, a strategic document Communicating Science in Slovenia <https://www.gov.si/novice/2024-08-02-komuniciranje-znanosti-v-sloveniji/> has been adopted in 2024, which defines key guidelines and activities in this field in Slovenia.

Part 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT

OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

72. 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

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73. 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)



Yes



No

74. 7.2 cont. **If yes**, please provide links as relevant and more information.

The formal basis that supports international collaboration in open science is the Resolution on the Slovenian Scientific Research and Innovation Strategy 2030 (ReZrIS30)
<https://www.gov.si/assets/ministrstva/MIZS/Dokumenti/ZNANOST/Nacionalni-dokumenti/Resolution-on-the-Slovenian-Scientific-Research-and-Innovation-Strategy-2030>
 containing Measure 6.2.1. that the national Open Science ecosystem must be coherent with international practices and the national open science structures and infrastructures are integrated into international networks and infrastructures. Additional national document that supports international collaboration in open science is Strategy for the internationalization of higher education and science in the Republic of Slovenia until 2030
<https://view.officeapps.live.com/op/view.aspx?src=https%3A%2F%2Fwww.gov.si%2Fassets%2Fministrstva%2FMVZI%2FVisoko-solstvo%2FDokumenti-VS%2FStrategija-internacionalizacije-VS-in-znanosti-2023.docx&wdOrigin=BROWSELINK>. Slovenia is promoting international collaboration on open science by supporting participation in international organisations, associations, forums and events. As for example "open science monitoring" including OSMI initiative <https://open-science-monitoring.org/>, RDA where we are financing operation of Slovenian RDA node <https://www.rd-alliance.org/groups/rda-slovenia/activity/>, besides, we are active in EOSC Association <https://eosc.eu/members/> with several delegates as well as in CoNOSC <https://conosc.org/membership/>. Besides, we have nominated national European Research Area Policy Agenda https://commission.europa.eu/system/files/2021-11/ec_rtd_era-policy-agenda-2021.pdf persons for open science actions (ERA action 1, ERA action 2 and ERA action 3). There are several other international and EU projects and activities that support open science with participants from Slovenija (CoARA <https://coara.eu/agreement/signatories/?category%5B0%5D=slovenia#signatories>, Knowledge Rights 21 <https://www.knowledgerights21.org/>, OPERAS <https://operas-eu.org/>, PALOMERA <https://operas-eu.org/projects/palomera/>, Mutual Learning Exercise on Research Careers <https://projects.research-and-innovation.ec.europa.eu/en/statistics/policy-support-facility/mutual-learning-exercise-research-careers>, European Citizen Science Association <https://www.ecsa.ngo/> etc.).

75. 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: (vii) 22.c)

☒ Yes

☐ No

76. 7.3 cont. **If yes**, please provide links as relevant and more information.

Slovenia is actively involved in the operation and development of the European Open Science Cloud Eosc, which supports open science at European Union level. Slovenia is also actively involved in the formation of policies to support the open science within the European Research Area.

Part 8. GENERAL CONSIDERATIONS

77. 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

☐ Yes

☒ No

78. ANY OTHER COMMENTS: If you wish, you may enter any additional comments or information here. You may also contact openscience@unesco.org

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